

25.10.25

Quiz 3
Wed 29.

Lazy Evaluation $\rightsquigarrow$ λ -expr

Prog Lang

Term

β

β
Input
output

--- $\rightarrow$
 n_1, n_2

0, 8
Numbers

+ * $\rightarrow$ defined fns

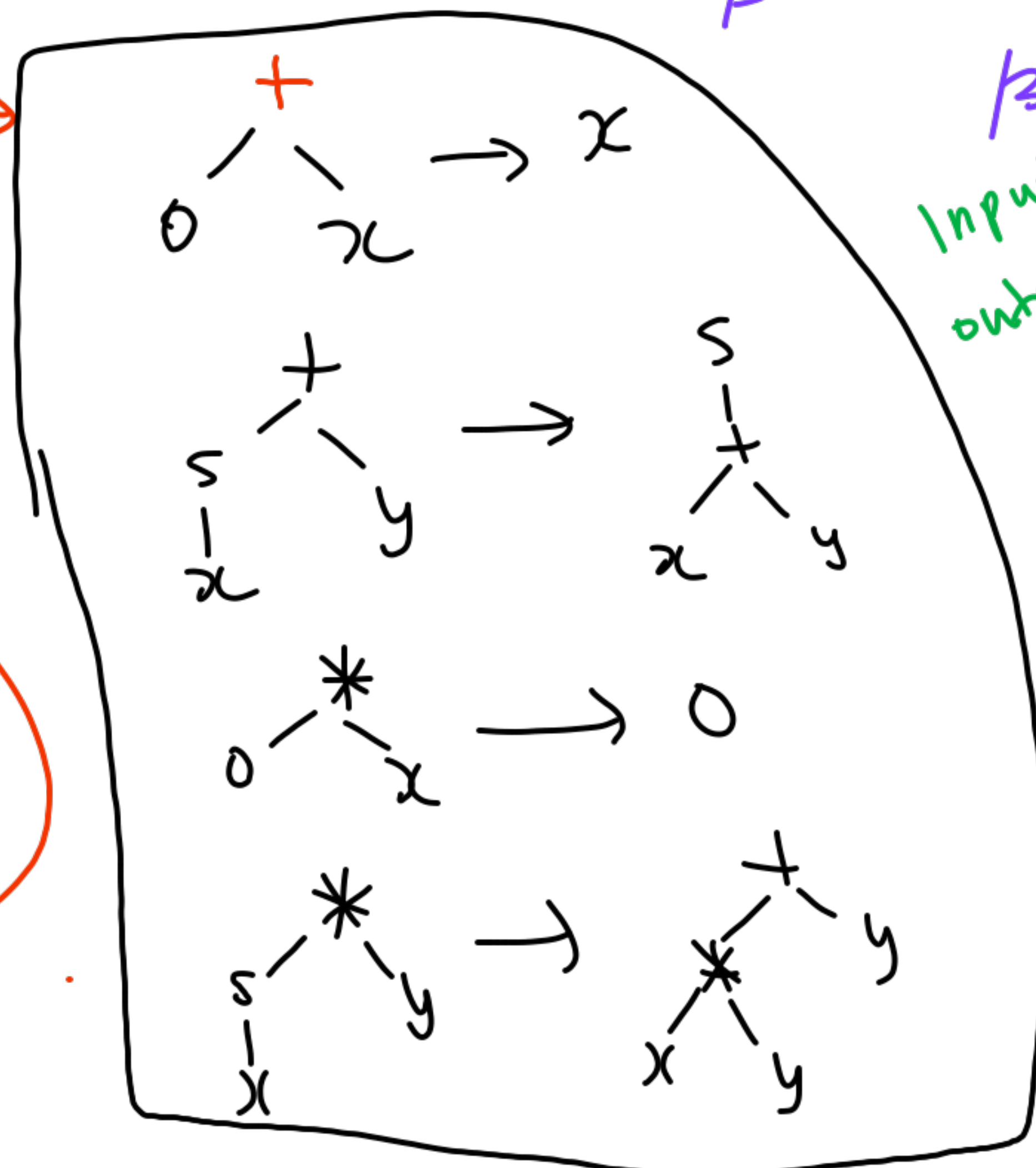
"Complete"

n_1 * $n_2 \rightarrow \dots \rightarrow n_3$

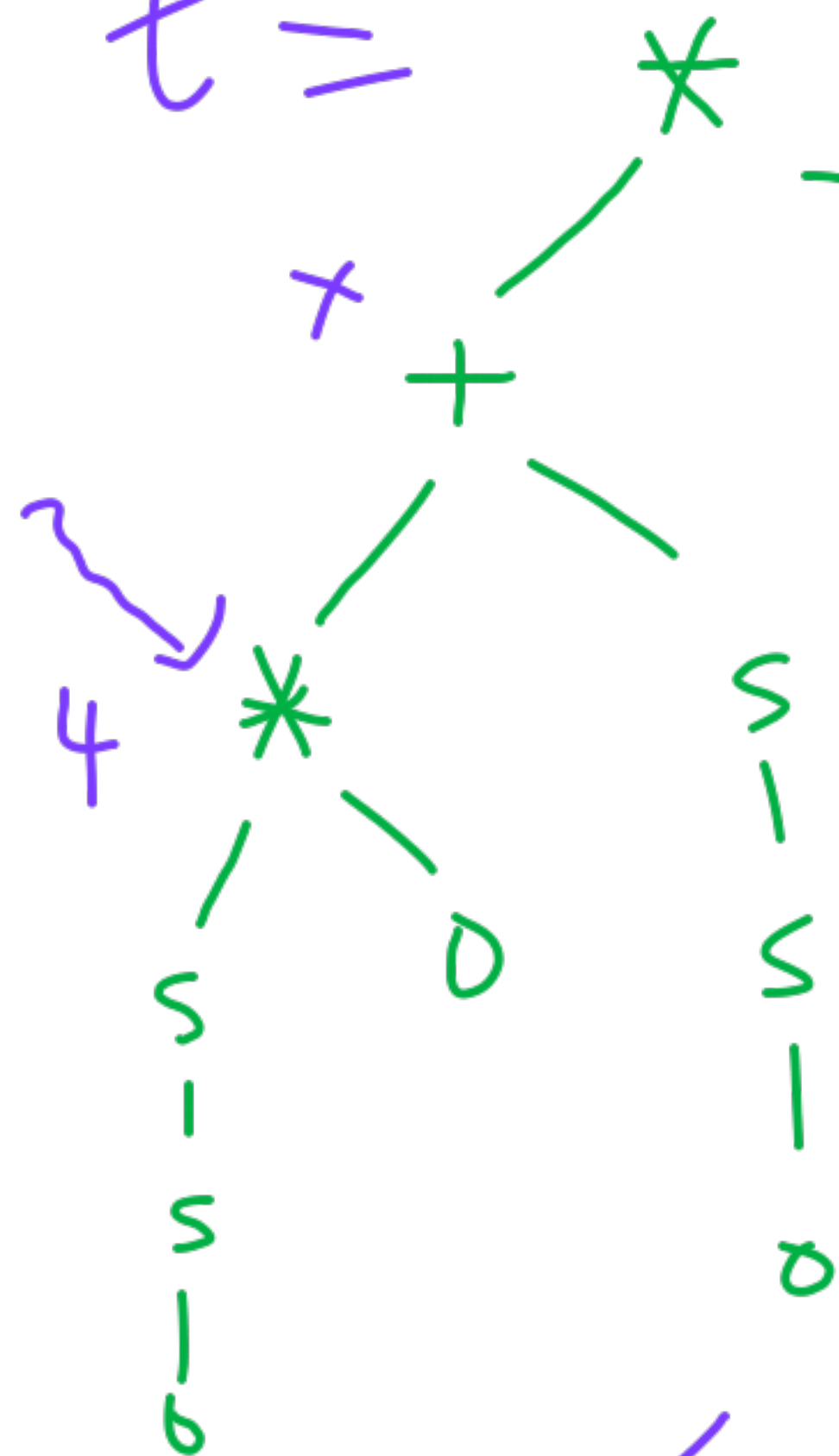
if-then-else
cond then else

def fact(x):
 if x = 0 then 1
 else x * fact(x-1)

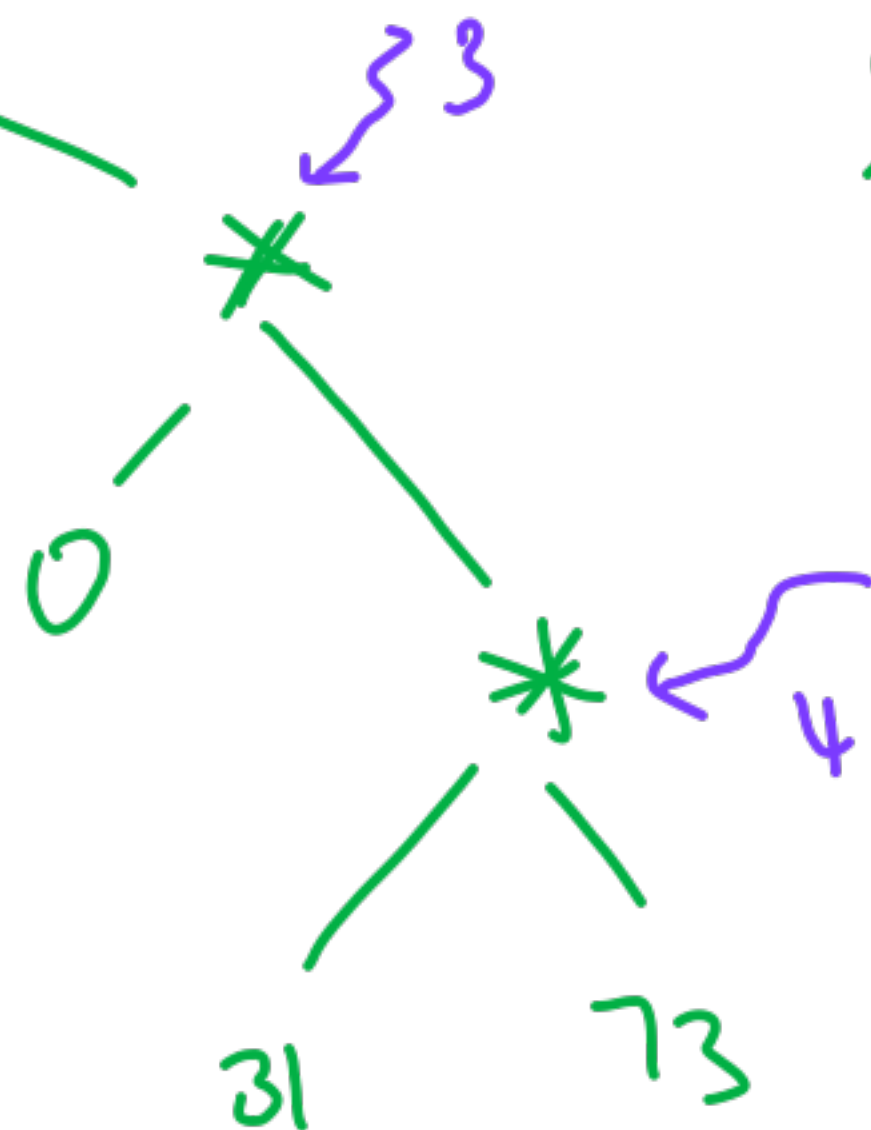
Spectre



t =



evaluate



Find its
Normal
form

no rule
applies

Redex

is a position
where a rule
can "apply"

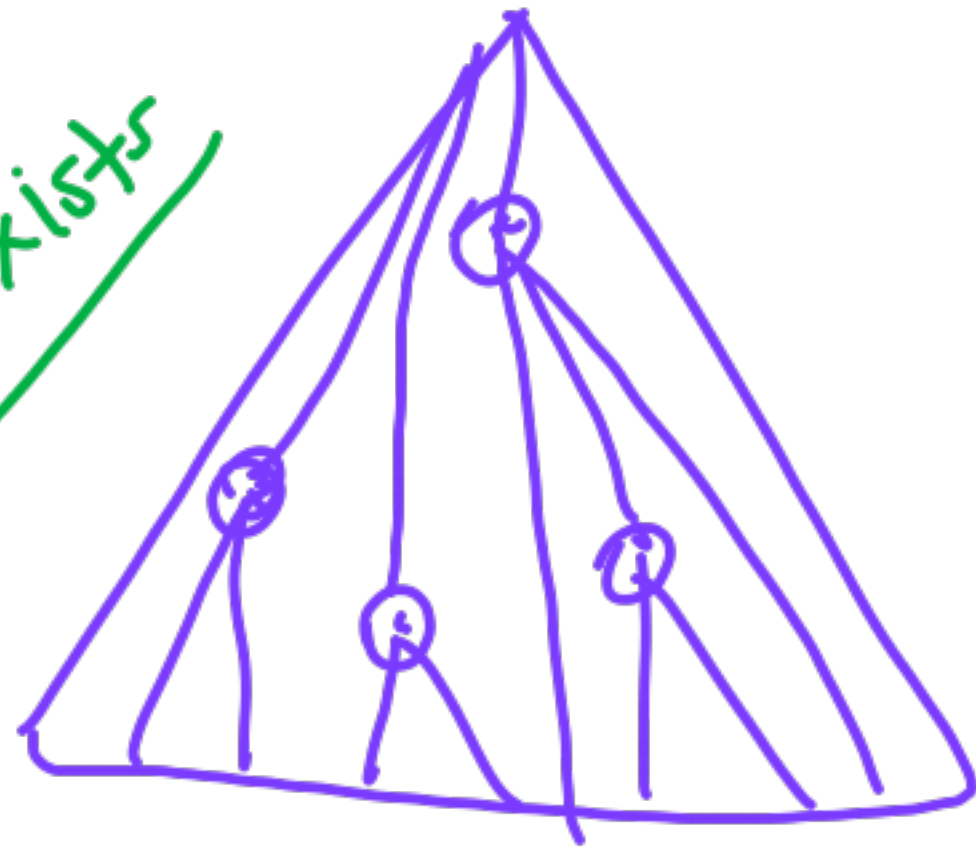
t_1

t_2

t_3

Strategies for evaluating a term

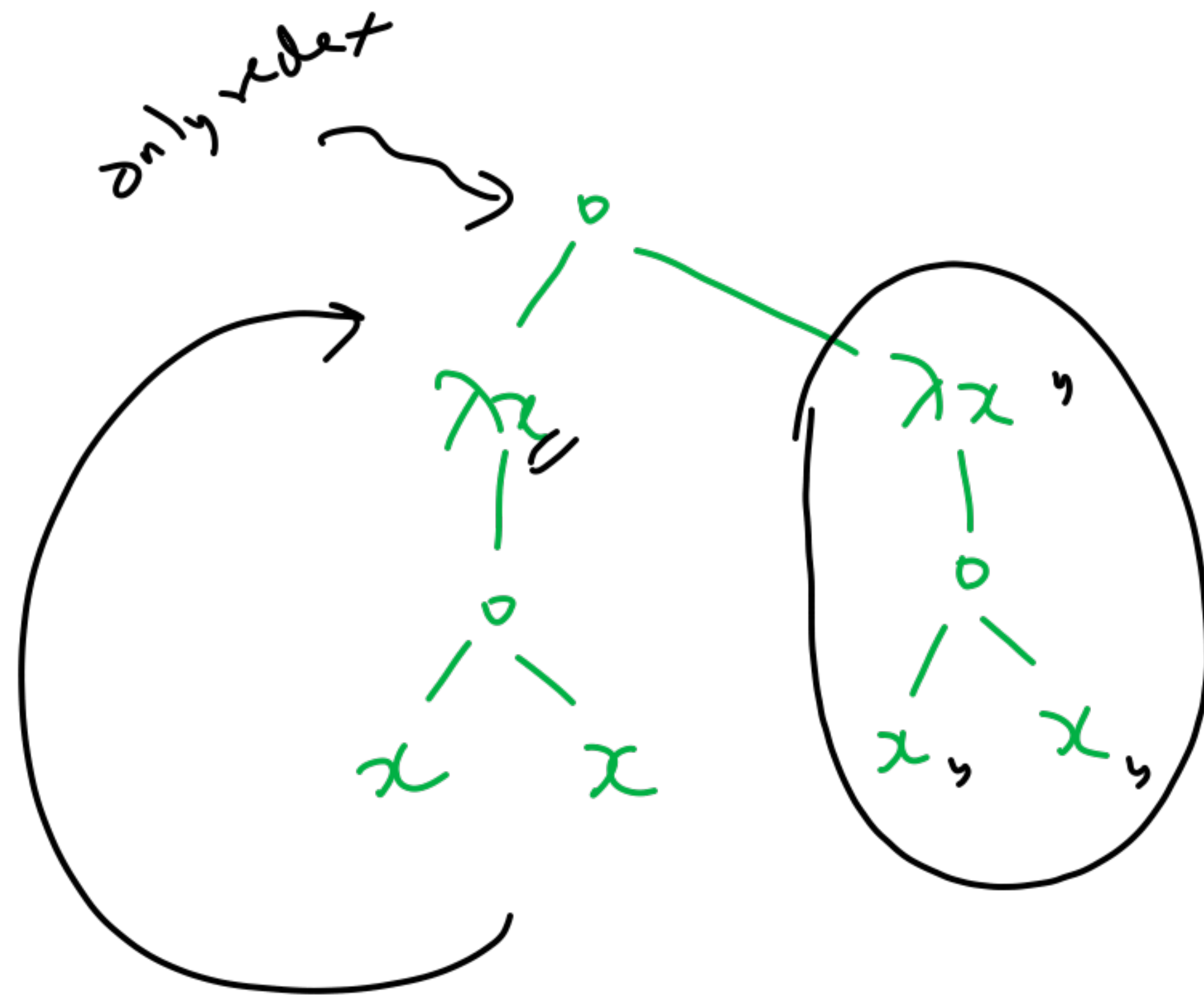
guaranteed
to find normal
form if it exists



Outermost
(Leftmost)
(Lazy)

Innermost $\times$
(bottom-most)
Eager (strict)

Mixed
(any)



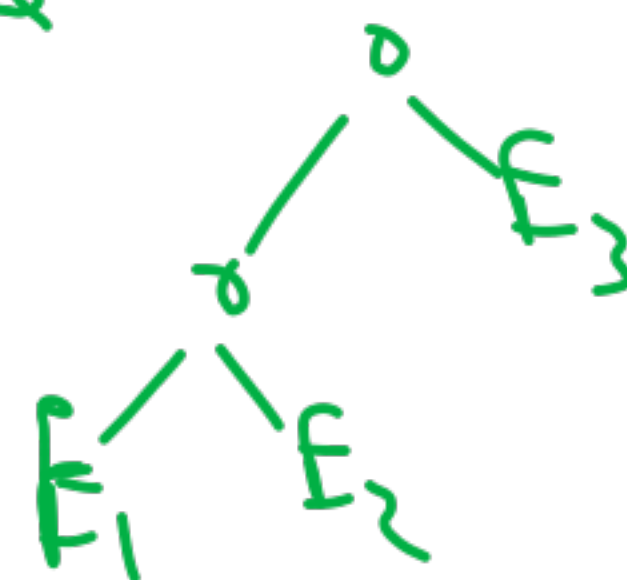
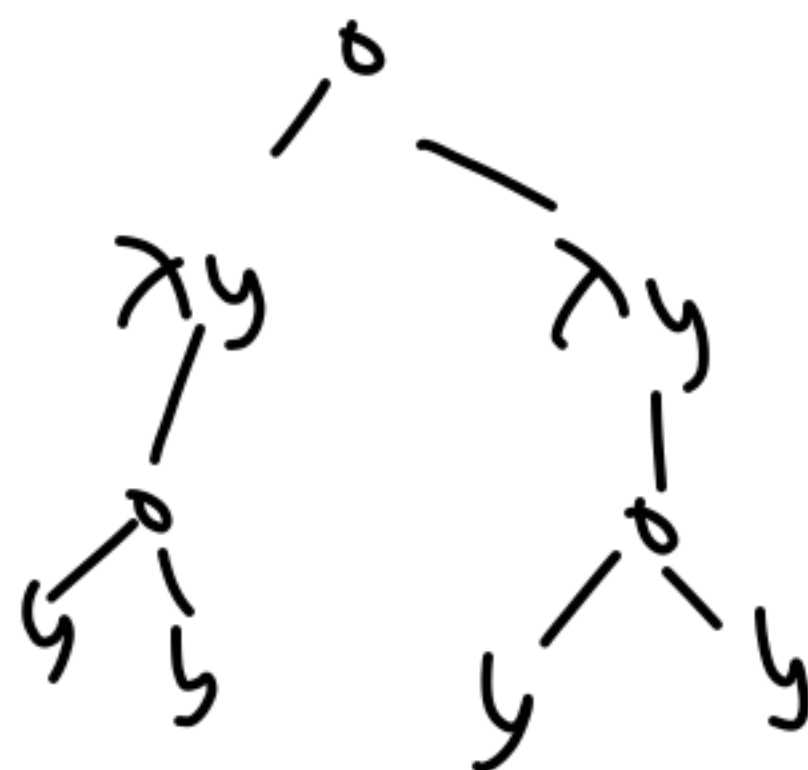
$$(\lambda x. x x) . (\lambda x x x)$$

$E_1 E_2 E_3$

$$((E_1 E_2) E_3)$$

left-associative

$$f(x) \rightarrow f(g(x))$$





$[]$ $[e : L]$



$[]$ $e \circ L$



meaningful term



"x is a member of L" → 0 not there
→ s if it is there
0

1. $\text{mem}(x, \overset{\text{nil}}{[]}) \rightarrow 0$

2. $\text{mem}(x, \underbrace{[x : y]}_{\text{cons}(x, y)}) \rightarrow \begin{matrix} s \\ | \\ 0 \end{matrix}$

3. $\text{mem}(x, \underbrace{[y : z]}_{\text{cons}(y, z)}) \rightarrow \text{mem}(x, z)$

$D \times \text{false}$
 $S \rightarrow \text{true}$
 (b)

① $\text{mem}(x, \text{nil}) \rightarrow 0$

② $\text{mem}(x, \text{cons}(y, z))$

$\rightarrow \text{ite}(\text{eq}(x, y), 0, \text{mem}(x, z))$

③ $\text{ite}(0, x, y) \rightarrow y$

④ $\text{ite}(s_0, x, y) \rightarrow x$

⑤ $\text{eq}(0, 0) \rightarrow s_0$ $\text{eq}(s_1, s_1) \rightarrow \text{eq}(x, y)$

⑥ $\text{eq}(0, s_1) \rightarrow 0$

⑦ $\text{eq}(s_1, 0) \rightarrow 0$

⑧

Tail-Recursion

